Decision Analytics (MGSC662) – Assignment3: NLP

Problem 1. The Markowitz Problem

A rational investor wishes for a low variance on return and a high-expected rate of return. However, the two goals are not aligned. For example, you can always keep variance down by investing in bonds over stocks, but you do so at the expense of a decent rate of return. So, an investor's optimal portfolio could be described by (r, σ), where r is a desired (and feasible) average rate of return, and σ^2 the minimal variance possible for this given r. (Put differently, an investor might wish to find the highest rate of return possible for a given acceptable level of risk.) This problem is due originally to Harry Markowitz in the 1950's.

The data in the Excel file "Markowitz" provide the performances of five stock exchange indices from 31 Dec 2010 to 1 June 2016.

(a) Formulate an NLP to devise the optimal portfolio that minimizes the portfolio risk subject to non-negative return. What are the optimal solution and value?
Hint: Notice that for the stock FTSE 100, for example, the rate of return from May 3 to June 1 2016 is (5,996.80 – 6230.80)/6230.80 = -0.0424.

(b) Assume the initial allocation is 20% in each index. Changing the position requires incurring transaction costs. Formulate an NLP to find the optimal portfolio to minimize risk subject to non-negative return. What are the optimal solution and value?

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Problem 2. The CYCOM Corporation Problem

John Brooks, managing director of the CYCOM Corporation, is trying to determine how to allocate CYCOM's research and development budget for the coming year. Six different projects are under consideration. John believes that the success of each and every project depends in part on the number of engineers assigned to the project. Let $x_1, \ldots, x_6$ denote the number of engineers assigned to projects 1,...,6. According to internal corporate estimates, the probability that each project will be successful is given according to the function in the third column of Table 1.

TABLE 1. The start-up cost of each project, the probability of success of each project, and the estimated profit of each project.

Project Number	*Start-up Cost (in $1,000)*	*Probability of Success*	*Profit if Successful (in $1,000)*
1	$325	$\frac{x_1}{x_1 + 1.1}$	$1,750
2	$200	$\frac{x_2}{x_2 + 0.5}$	$700
3	$490	$\frac{x_3}{x_3 + 2.5}$	$1,300
4	$125	$\frac{x_4}{x_4 + 1.6}$	$800
5	$710	$\frac{x_5}{x_5 + 2.2}$	$1,450
6	$240	$\frac{x_6}{x_6 + 2.4}$	$1,300

For example, according to Table 1, the probability that project 3 will be successful is

$$p_3 = \frac{x_3}{x_3 + 2.5}$$

If John were to assign $x_3 = 5$ engineers to project 3, then the probability of success of the project would be $p_3 = \frac{5}{5+2.5} = 0.667$.

Table 1 also shows the start-up cost of each project and the estimated profit of each project if it is successful.

Cycom can assign up to 25 engineers to these six projects. The cost of an engineer is $150,000 (per year). An engineer can be assigned to more than one project. For example, $x_2 = 3.4$ means that 3.4 engineers are assigned to project 2, which means that 3 engineers are assigned to this project full-time and one engineer is assigned to this project only 40% of the time.

If a project is unsuccessful, it incurs its full start-up cost, shown in the second column of Table 1. If the project is successful, it realizes the profit value shown in the fourth column of Table 1. This value includes the start-up cost of the project (but does not include the cost of engineers). Thus, for example, if project 1 is unsuccessful, it incurs a cost of $325,000. If project 1 is successful, it realizes a profit of $1,750,000.

(a) Construct and solve a nonlinear optimization model to determine the number of engineers to assign to each project that will maximize the expected contribution to profit of the six projects

minus the cost of assigning engineers to the projects and the start-up costs of the projects. What is the optimal solution? (Rounded to one decimal place)

(b) Construct and solve an optimization model that minimizes the standard deviation of the contribution to profit subject to the constraint that the expected contribution to profit (minus the cost of engineers) is at least $1.1 million. What is the optimal solution? (Rounded to one decimal place)

Hint-1: Assume that the success or not of one project is statistically independent of the success or not of any other project.

Hint-2: If random variable X takes value A with probability p and value B with probability (1-p), then:

$$E(X) = p.A + (1-p).B$$

$$Var(X) = (A-B)^2.p.(1-p)$$